

Intellectual Merit

Frames, like ortho-normal bases, give a continuous, linear, and stable reconstruction formula for vectors in a Hilbert space. However, frames allow for redundancy, and this makes frames much more adaptable for theory and applications. Interest in frame theory took off in 1986 when Daubechies, Grossmann, and Meyer initiated the use of frames in signal processing. Since then, frame theory has grown into a rich mathematical area replete with both interesting theoretical questions as well as important applications in physics and engineering. Our proposal focuses on the specific application of frame theory to phase retrieval. Given a frame $(x_j)_{j \in J}$ in a real Hilbert space H , the goal of phase retrieval is to recover a vector $x \in H$ (or $-x$) from the absolute value of the frame coefficients $(|\langle x, x_j \rangle|)_{j \in J}$. Phase retrieval's most famous application was in X-ray crystallography and was fundamental to uncovering the double helix structure of DNA. Since then, phase retrieval has become critical to coherent diffractive imaging and speech recognition technology among many other important applications. Because of its broad utility, much of the research concerning phase retrieval has focused on its use and application. In contrast, many interesting questions concerning its mathematical foundations remain open. We propose a unique approach to building on the mathematical theory of phase retrieval by using an innovative combination of techniques from frame theory and the geometry of Banach spaces. The PI's expertise is largely in the geometry of Banach spaces and infinite dimensional frame theory, and the Co-PI's expertise is largely in finite frame theory and phase retrieval. By working together, we are able to offer new approaches to long outstanding problems in frame theory as well as to extend the fundamentals of frame theory and phase retrieval in new directions.

We propose six separate projects on frame theory, and each project includes multiple open questions and avenues for research. We now briefly introduce each topic and discuss their intellectual merit.

(1) Minimal structures for phase retrieval

Phase retrieval with a basis is impossible, and it requires the redundancy of a frame. This naturally leads to the question of just how much redundancy is necessary to do phase retrieval. It is known that a frame of $2n - 1$ vectors is needed to do phase retrieval in $\mathbb{R}^n$. This bound can be lowered when one uses a fusion frame instead of a frame. When using a fusion frame for phase retrieval we use projection operators composed with the norm instead of rank-1 linear measurements composed with absolute value. However, necessary bounds for this procedure are known for only very specific values of n [Ed15]. There is a lot of flexibility in this situation as one can choose both the dimensions and arrangements of the projections. Because of this, determining the necessary bounds for phase retrieval by projections is a mathematically interesting problem which can provide insight in determining optimal configurations of phase retrieval in application.

(2) Stable phase retrieval

Given a measure space (M, μ) , we propose a formulation of phase retrieval for subspaces of $L_2(M)$. In the case that μ is counting measure on M , this coincides with the standard case of phase retrieval using a frame. The advantage of our formulation is that it allows for stable phase retrieval for infinite dimensional Hilbert spaces, which was previously considered to be impossible. This new concept opens the door to many new questions. Furthermore, it provides new tools for analyzing phase retrieval in finite dimensions. One can realize certain subspaces of $L_2(M)$ as limits in some sense of random frames for finite dimensional Hilbert spaces, and thus information about those subspaces could be applied to the random frames.

(3) Continuous Schauder frames in Banach spaces and phase retrieval

A collection of vectors $(x_j)_{j \in J}$ in a Hilbert space H is a frame if and only if H may be embedded into a larger Hilbert space with a Riesz basis which may be projected down to $(x_j)_{j \in J}$. Likewise, Schauder frames for Banach spaces can be characterized as projections of Schauder bases. This does not work for continuous frames as there is no such thing as a continuous basis over a non-atomic measure space. However, we propose using embeddings into Banach function spaces as a suitable replacement for this when using continuous Schauder frames. This will allow us to extend many of the fundamental properties for discrete coordinate systems in Banach spaces to the continuous setting. Furthermore, this more general setting for continuous Schauder frames allows for constructions which are both impossible for continuous frames for Hilbert spaces as well as impossible for discrete Schauder frames. For example, phase retrieval using a continuous frame for an infinite dimensional Hilbert space is always unstable, but we show that it is possible to do stable phase retrieval using a

continuous Schauder frame. The benefit of extending phase retrieval to the Banach lattice setting is twofold. The rich theory of Banach lattices provides many tools for us to use in studying phase retrieval, and by introducing phase retrieval into this setting we open a new avenue of research in Banach lattices.

(4) Sampling continuous frames and extensions of the Feichtinger conjecture

Every bounded continuous frame for a Hilbert space may be sampled at countably many points to get a discrete frame [FS19]. We now consider continuous frames with some desirable structure and question when one could sample the continuous frame to obtain a discrete frame which shares that same structure. We consider this in the context of localized continuous frames, and continuous frames which do phase retrieval.

(5) Coordinate systems with constrained structure

Given $f \in L_2(\mathbb{R})$, the closed span of the sequence of integer translates $\overline{\text{span}}(T_n f)_{f \in \mathbb{Z}}$ is called a principal shift invariant subspace, and it plays an important role in approximation theory and applied harmonic analysis. A simple Fourier transform argument shows that $\overline{\text{span}}(T_n f)_{f \in \mathbb{Z}}$ can never be all of $L_2(\mathbb{R})$. Remarkably, if $(\lambda_n)_{n \in \mathbb{Z}} \subseteq \mathbb{R} \setminus \mathbb{Z}$ and $|\lambda_n - n| \rightarrow 0$ then there exists $f \in L_2(\mathbb{R})$ such that the span of $(T_{\lambda_n} f)_{n \in \mathbb{Z}}$ is dense in $L_2(\mathbb{R})$. Though it is possible for a sequence of translates of a single function to be dense in $L_2(\mathbb{R})$, such a sequence will never be a Riesz basis or a frame for $L_2(\mathbb{R})$. The most significant open question in this area is whether or not there exists a sequence of translates of a single function which is a conditional Schauder basis for $L_2(\mathbb{R})$. There has been little progress towards answering this question, but recently the PI solved a related long outstanding problem by constructing a conditional Schauder basis for $L_2(\mathbb{R})$ consisting of only non-negative functions [FPT20]. We propose to adapt this new idea with other methods established in [FOSZ14] to construct a basic sequence of translations of a single function for $L_2(\mathbb{R})$.

(6) Phase retrieval for vector bundles.

Instead of studying a frame for a fixed Hilbert space, we now consider a frame for the tangent space of a manifold which moves smoothly on the manifold. Moving bases have long been studied for manifolds, but they do not exist for many important manifolds. In contrast to this, one can use the Whitney immersion theorem to show that every n -dimensional Riemannian manifold has a moving frame for its tangent space of $2n - 1$ vectors. We propose developing the theory of how phase retrieval may be done in this continuous setting. That is, instead of recovering a single vector up to a phase factor from the absolute values of its frame coefficients, our goal is to use a continuously moving frame to recover a vector field up to an equivalence relation. Surprisingly, many of the fundamental results for phase retrieval no longer for vector bundles. For example, it is no longer the case that a frame for a finite dimensional space does phase retrieval if and only if it does stable phase retrieval, and this fails even for the simple case of the tangent bundle for S^2 . Thus developing a theory of phase retrieval for sections of vector bundles will be both interesting and non-trivial.

1. MINIMAL STRUCTURES FOR PHASE RETRIEVAL

A collection of vectors $(\phi_j)_{j \in J}$ in a Hilbert space H is called a *frame* if there are constants $0 < A \leq B < \infty$ so that for all $x \in H$, $A\|x\|^2 \leq \sum_{j \in J} |\langle x, \phi_j \rangle|^2 \leq B\|x\|^2$. The constants A and B are called a *lower frame bound* and *upper frame bound* respectively. A frame is called *A-tight* if the frame bounds can be chosen to satisfy $A = B$ and is called *Parseval* if $A = B = 1$. The *analysis operator* for a frame $(\phi_j)_{j \in J}$ is the map $T : H \rightarrow \ell_2(J)$ given by $Tx = (\langle x, \phi_j \rangle)_{j \in J} \in \ell_2(J)$. Note that $(\phi_j)_{j \in J}$ has lower frame bound A and upper frame bound B can be equivalently stated as $A\|x\|^2 \leq \|Tx\|_{\ell_2(J)}^2 \leq B\|x\|^2$ for all $x \in H$. If T is the analysis operator for $(\phi_j)_{j \in J}$ then the *frame operator* for $(\phi_j)_{j \in J}$ is the map $S = T^*T : H \rightarrow H$. That is, $Sx = \sum_{j \in J} \langle x, \phi_j \rangle \phi_j$ for all $x \in H$. We then have that $(\phi_j)_{j \in J}$ is a frame of H with lower frame bound A and upper frame bound B if and only if the frame operator S is bounded, invertible, and satisfies $\|S\| \leq B$ and $\|S^{-1}\| \leq A^{-1}$. In the case that $(\phi_j)_{j \in J}$ is Parseval, then the frame operator is the identity and hence $x = \sum_{j \in J} \langle x, \phi_j \rangle \phi_j$ for all $x \in H$. Thus Parseval frames, like ortho-normal bases, satisfy Parseval's identity.

Frames play similar roles in applications to ortho-normal bases, but frames allow for redundancy. Not only does this redundancy make frames more resilient to error, but the redundancy of a frame is necessary for phase retrieval. That is, it is impossible to do phase retrieval with a basis. The goal of phase retrieval is to recover a vector (up to a unimodular scalar) from a collection of magnitudes of linear measurements.

More specifically, if $(\phi_j)_{j \in J}$ is a frame of a Hilbert space H then we say that $(\phi_j)_{j \in J}$ does *phase retrieval* if whenever $x, y \in H$ satisfy $|\langle x, \phi_j \rangle| = |\langle y, \phi_j \rangle|$ for all $j \in J$ there exists a scalar θ with $|\theta| = 1$ so that $x = \theta y$. There are many applications in physics or engineering where one is only able to obtain the magnitude of a linear measurement such as in speech recognition [BR99], coherent diffraction imaging [MCKS99], X-ray crystallography [T03], and transmission electron microscopy [K98] and phase retrieval plays a critical role in those applications among many others.

Using rank one linear measurements for phase retrieval is done with a frame, where as higher rank linear measurements would be done with a fusion frame. A collection of scalars $(a_j)_{j \in J}$ and subspaces $(H_j)_{j \in J}$ is called a *fusion frame* of H if there exists constants $0 \leq A \leq B < \infty$ such that $A\|x\|^2 \leq \sum_{j \in J} |a_j|^2 \|P_j x\|^2 \leq B\|x\|^2$ for all $x \in H$ where P_j is orthogonal projection onto H_j . We say that the fusion frame does *phase retrieval by projection* or that the subspaces $(H_j)_{j \in J}$ do *phase retrieval by projection* if whenever $x, y \in H$ satisfy $\|P_j x\| = \|P_j y\|$ for all $j \in J$ there exists a scalar $|\theta| = 1$ so that $x = \theta y$. Recall that redundancy is necessary for phase retrieval to be possible, and this leads one to consider just how much redundancy is necessary. When doing phase retrieval with a frame for $\mathbb{R}^n$ it is necessary that the frame contain at least $2n - 1$ vectors, and furthermore a generic frame of $2n - 1$ vectors does phase retrieval in $\mathbb{R}^n$ [BCE06]. It was assumed that the same bound of $2n - 1$ would hold when doing phase retrieval by projections, but in [Xu17] the authors gave an example of six 2-dimensional subspaces which did phase retrieval in $\mathbb{R}^4$.

Problem 1.1. For each $n \in \mathbb{N}$, what is the minimum number of subspaces which can do phase retrieval by projection in $\mathbb{R}^n$?

A frame $\Phi = \{\phi_i\}_{i=1}^k$ in $\mathbb{R}^n$ is said to satisfy the *complement property* if for all subsets $I \subset [k]$, either $\text{span}\{\phi_i\}_{i \in I} = \mathbb{R}^n$ or $\text{span}\{\phi_i\}_{i \in I^c} = \mathbb{R}^n$. The complement property is very useful as it gives a discrete characterization of phase retrieval. That is, a frame does phase retrieval for $\mathbb{R}^n$ if and only if the frame satisfies the complement property [BCE06]. This can be extended to phase retrieval by projections in the following way. A family of projections $\{P_j\}_{j=1}^m$ in $\mathbb{R}^n$ does phase retrieval if and only if for every $0 \neq x \in \mathbb{R}^n$, the vectors $\{P_j x\}_{j=1}^m$ span the space [Ed15]. This result is useful when studying phase retrieval by projections, but in order to solve Problem 1.1 we would like to have a discrete version. One possible way to discretize the property is to consider the lattice of subspaces Λ of $\mathbb{R}^n$ generated by a collection of subspaces $(H_j)_{j=1}^m$ under the operations of intersection and span. One could then try to determine whether or not the subspaces $(H_j)_{j=1}^m$ do phase retrieval based on the structure of the discrete collection of subspaces Λ . Another approach, as taken by the Co-PI in [CGJT17], is to consider a weakening of the phase retrieval condition which would allow us to reduce the complexity of the problem. One could then attack the weakened problem, and then determine situations when the weak condition is equivalent to phase retrieval.

Problem 1.2. Is there a discrete version of the complement property for phase retrieval by projections? Can this be done for a weakened version of phase retrieval by projections?

An interesting aspect of phase retrieval by projections is that one can consider how the ranks of the projections must relate to the dimension of the space. One of the known results is that if a family of rank $n - 1$ projections does phase retrieval then the family must include at least $2n - 2$ projections.

Problem 1.3. Given $k, n \in \mathbb{N}$, what are the possible ranks for which there exists a sequence of projections $(P_i)_{i=1}^k$ which satisfies phase retrieval in $\mathbb{R}^n$.

The problem of determining minimal structures for phase retrieval in $\mathbb{C}^n$ is open even for phase retrieval using frames. In [BoHa] there are some constructions of vectors satisfying phase retrieval in $\mathbb{C}^n$ for certain values of n , and it is shown that for all $n \geq 2$, phase retrieval can be done in $\mathbb{C}^n$ with $4n - 4$ vectors. It is conjectured in [CoEdHe15] that a generic frame of $4n - 4$ vectors would do phase retrieval in $\mathbb{C}^n$ and not fewer vectors can do phase retrieval. However there is an example in [V15] showing that 11 vectors can satisfy phase retrieval in $\mathbb{C}^4$. The main question we can ask here is that:

Problem 1.4. Given $n \in \mathbb{N}$, what is the minimum number of vectors required for a frame to do phase retrieval in $\mathbb{C}^n$? What is the minimum number of subspaces required to do phase retrieval by projection in $\mathbb{C}^n$?

In looking at possible ways to answer problem 1.1 and problem 1.4, we can ask how it would be possible to modify the dimensions of a family of subspaces which do phase retrieval by projection. This would allow us to greatly simplify problems in phase retrieval by projection.

Problem 1.5. Suppose a family of subspaces $(W_j)_{j=1}^m$ does phase retrieval and $\text{codim}(W_k) \geq 2$ for some $1 \leq k \leq m$. Then does there exist a vector $v \in W_k^\perp$ so that $\text{span}\{W_k, v\} \cup \{W_i\}_{i \neq k}$ does phase retrieval? What are necessary and sufficient conditions for this to be possible?

Notice that if it is always possible to modify subspaces in such a way then by repeating the procedure we would end up with a family of hyperplanes which do phase retrieval. Thus in $\mathbb{R}^n$ we would have that the minimum number of subspaces which do phase retrieval by projection would be at least $2n - 2$.

When talking about modifying a subspace by adding or removing a vector, it is important to know what properties can be transferred to the complement space. If $\{W_i\}_{i=1}^m$ does phase retrieval, does $\{W_i^\perp\}_{i=1}^m$ do phase retrieval as well? It is shown in [CCPW16] that this is not true in general. Norm retrieval is the property needed to pass phase retrieval to orthogonal complements. We say that a collection of subspaces $\{W_i\}_{i=1}^m$ (or $\{P_i\}_{i=1}^m$) yields *norm retrieval* if for all $x, y \in \mathbb{H}^n$ satisfying $\|P_i x\| = \|P_i y\|$ for all $i = 1, 2, \dots, m$ then $\|x\| = \|y\|$. If a family of subspaces satisfy phase retrieval, then it satisfies norm retrieval if and only if the complement space does phase retrieval. In the previous work [CGJT17], the Co-PI has proven the following theorem:

Theorem 1.6. Let $\{W_i\}_{i=1}^m$ be a family of subspaces of $\mathbb{R}^n$ that satisfies norm retrieval. Then whenever one decomposes $W_i = W_i^1 \oplus W_i^2$ for some $i \in [m]$, we have that if $x \perp W_i^1$ and $y \perp W_i^2$ then $x \perp y$.

By incorporating norm retrieval into phase retrieval by projection we hope, we hope to classify the subspaces which do phase retrieval by projection after growing or shrinking a subspace by one dimension.

2. STABLE PHASE RETRIEVAL IN HILBERT SPACES

Let H be a separable Hilbert space and $(\phi_j)_{j \in J} \subseteq H$ be a countable collection of vectors. Given $f \in H$, the goal of phase retrieval is to determine f (up to a unimodular scalar) from the collection of values $(|\langle f, \phi_j \rangle|)_{j \in J}$. In particular, when H is a real Hilbert space we say that $(\phi_j)_{j \in J}$ does *phase retrieval* if whenever $f, g \in H$ and $(|\langle f, \phi_j \rangle|)_{j \in J} = (|\langle g, \phi_j \rangle|)_{j \in J}$ we have that $f = g$ or $f = -g$.

In the case that $(\phi_j)_{j \in J}$ is a frame of H , the map $f \mapsto (\langle f, \phi_j \rangle)_{j \in J}$ is an isomorphic embedding of H into $\ell_2(J)$. Thus, we may consider H as a subspace of $\ell_2(J)$. The phase retrieval problem is now equivalent to recovering $f = (f_j)_{j \in J} \in H \subseteq \ell_2(J)$ (up to a unimodular scalar) from its absolute value $|f| := (|f_j|)_{j \in J} \in \ell_2(J)$. This equivalent formulation leads us to the definition that a real Hilbert space $H \subseteq \ell_2(J)$ does *phase retrieval* if for all $f, g \in H \subseteq \ell_2(J)$ with $|f| = |g|$ we have that $f = g$ or $f = -g$.

Any use of phase retrieval in application will involve some error. Thus it is important for phase retrieval to not only be possible, but for it also to be stable. We say that $H \subseteq \ell_2(J)$ does *stable phase retrieval* if there exists a constant $C > 0$ such that for all $f, g \in H$ we have that $\min(\|f - g\|, \|f + g\|) \leq C\||f| - |g|\|$. Phase retrieval is always stable in finite dimensions. That is, if $H \subseteq \ell_2(J)$ is finite dimensional and does phase retrieval then $H \subseteq \ell_2(J)$ does stable phase retrieval [B17][CCPW13]. In contrast to this, phase retrieval is unstable for every infinite dimensional subspace of ℓ_2 [CCD16]. We propose to consider phase retrieval for infinite dimensional subspaces of $L_2(\mathbb{R})$ instead of ℓ_2 . Though $L_2(\mathbb{R})$ and ℓ_2 are isometric as Hilbert spaces, their structure as Banach lattices is very different. In joint work with Robert Calderbank, Ingrid Daubechies, and Nikki Freeman, the PI has proven that there exists an infinite dimensional closed subspace $H \subseteq L_2(\mathbb{R})$ and constant $C > 0$ such that for all $f, g \in H$ we have that $\min(\|f - g\|_{L_2}, \|f + g\|_{L_2}) \leq C\||f| - |g|\|_{L_2}$ [CDFF20]. That is, unlike in ℓ_2 , stable phase retrieval is possible for infinite-dimensional subspaces of $L_2(\mathbb{R})$. In particular, we prove the following theorem.

Theorem 2.1. Let $(y_j)_{j=1}^\infty$ be an ortho-normal sequence of mean-zero independent random vectors in $L_2([0, 1])$. Let $x_j = y_j + 1_{(j, j+1]}$ for all $j \in \mathbb{N}$. Then phase retrieval is stable for the closed subspace $\overline{\text{span}}(x_j)$ of $L_2(\mathbb{R})$ if and only if there exists a constant $A > 0$ such that $\|y_j\|_{L_1([0, 1])} \geq A$ for all $j \in \mathbb{N}$. In particular, phase retrieval is stable for $\overline{\text{span}}(x_j)$ whenever $(y_j)_{j=1}^\infty$ is an iid sequence of random vectors with finite second moment.

Thus, we have a complete classification of when phase retrieval is stable for a subspace of $L_2(\mathbb{R})$ of the form $\overline{\text{span}}(y_j + 1_{(j, j+1]})$ where $(y_j)_{j=1}^\infty$ is an ortho-normal sequence of mean-zero independent random variables in $L_2([0, 1])$. Furthermore, the classification only depends on an easily verified property of the sequence $(y_j)_{j=1}^\infty$. The reason for adding the indicator functions $1_{(j, j+1]}$ is that it is a simple addition which allows us to easily distinguish $|y_j + 1_{(j, j+1]}|$ from $|y_i + 1_{(i, i+1]}|$ for $i \neq j$. Indeed, if $(y_j)_{j=1}^\infty = (R_j)_{j=1}^\infty$ is the sequence

of Rademacher functions (independent symmetric ± 1 random variables) then $|R_j| = |R_i| = 1$ for all $i, j \in \mathbb{N}$ and so the subspace $\overline{\text{span}}(R_j)$ does not do phase retrieval, where as the subspace $\overline{\text{span}}(R_j + 1_{(j,j+1]})$ does stable phase retrieval. This naturally leads to the problem of determining a property of $(y_j)_{j=1}^\infty$ which allows us to obtain stable phase retrieval without adding on the indicator functions.

Problem 2.2. Let $(y_j)_{j=1}^\infty$ be an ortho-normal sequence of mean-zero independent random variables in $L_2([0, 1])$. Can we give a simple characterization of when phase retrieval is stable for the closed subspace $\overline{\text{span}}(y_j)$ of $L_2([0, 1])$? In particular, is phase retrieval stable for $\overline{\text{span}}(y_j)$ of $L_2([0, 1])$ if and only if there exist constants $0 < A \leq B < 1$ such that $A \leq \|y_j\|_{L_1([0,1])} \leq B$ for all but at most one $j \in \mathbb{N}$?

Note that the condition that there exist constants $0 < A \leq B < 1$ such that $A \leq \|y_j\|_{L_1([0,1])} \leq B$ for all but at most one $j \in \mathbb{N}$ is a necessary condition for the subspace $\overline{\text{span}}(y_j)$ to do stable phase retrieval in $L_2([0, 1])$. If there did not exist a constant A with $A \leq \|y_j\|_{L_1([0,1])}$ for all $j \in \mathbb{N}$ then we could find a subsequence of $(y_j)_{j=1}^\infty$ whose support is almost disjoint and then use the arguments in [CCD16] or [AG17] to prove that phase retrieval is not stable. If there does not exist such a $B < 1$ then because a mean-zero random variable f has Radamacher distribution if and only if $\|f\|_{L_2} = \|f\|_{L_1}$ we can conclude that $\overline{\text{span}}(y_j)$ does not do stable phase retrieval for the same reason that $\overline{\text{span}}(R_j)$ does not do phase retrieval.

All of our constructions and proofs have so far relied heavily on probabilistic techniques. We propose a different harmonic analysis approach for obtaining stable phase retrieval in the following problem.

Problem 2.3. For all $j \in \mathbb{N}$, let $y_j : [0, 1] \rightarrow \mathbb{R}$ be the function $y_j(t) = \sin(2^j \pi t)$. Then is phase retrieval stable for the closed subspace $\overline{\text{span}}(y_j + 1_{(j,j+1]})$ of $L_2(\mathbb{R})$? Is phase retrieval stable for the closed subspace $\overline{\text{span}}(y_j)$ of $L_2(\mathbb{R})$?

By restricting to finite-dimensional subspaces, our proposed research on stability of phase retrieval in infinite dimensions will give new constructions where phase retrieval is uniformly stable in finite dimensions with stability constant independent of the dimension. Doing phase retrieval for $\overline{\text{span}}(y_j + 1_{(j,j+1]})_{j=1}^\infty \subseteq L_2(\mathbb{R})$ obviously requires evaluation at infinitely many points. On the other hand, for each $n \in \mathbb{N}$, we would like to obtain uniform stability for phase retrieval for $\text{span}(y_j + 1_{(j,j+1]})_{j=1}^n \subseteq L_2(\mathbb{R})$ by evaluating at only finitely many points. In [CDDF20], we consider the case that $(y_j)_{j=1}^\infty$ is an ortho-normal sequence of mean-zero independent random vectors in $L_2([0, 1])$ and $\|y_j\|_{L_1([0,1])} \geq A$ for all $j \in \mathbb{N}$ for some $A > 0$. Given $n \in \mathbb{N}$ and $x \in \text{span}(y_j + 1_{(j,j+1]})_{j=1}^n \subseteq L_2(\mathbb{R})$ we evaluate x at one point in each interval $[j, j+1]$ for all $1 \leq j \leq n$, and we evaluate x at a total of m points in $[0, 1]$ where m depends on the dimension n . We prove that if m on the order of $n \log(n)$ points are sampled independently and uniformly in $[0, 1]$ then we obtain uniformly stable phase retrieval with high probability. Furthermore, if the sequence $(y_j)_{j=1}^\infty$ is uniformly sub-Gaussian then we obtain uniform stability with high probability by only evaluating at m on the order of n points in $[0, 1]$. We propose the following finite dimensional extension of Problem 2.2.

Problem 2.4. Let $(y_j)_{j=1}^\infty$ be an ortho-normal sequence of mean-zero independent random variables in $L_2([0, 1])$ such that phase retrieval is stable for $\overline{\text{span}}(y_j)_{j=1}^\infty \subseteq L_2([0, 1])$. Does there exist a constant $C > 0$ and function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all $n \in \mathbb{N}$ if $m = f(n)$ points $(t_j)_{j=1}^m$ are sampled independently and uniformly in $[0, 1]$ then with high probability, we have that

$$(1) \quad \|x - y\|_{L_2([0,1])} \leq C m^{-1/2} \left(\sum_{j=1}^m |x(t_j)| - |y(t_j)| \right)^2)^{1/2} \quad \text{for all } x, y \in \text{span}(y_j)_{j=1}^n?$$

How can we construct upper bounds on the order of $m = f(n)$? May $f(n)$ be chosen on the order of $n \log(n)$? If $(y_j)_{j=1}^\infty$ are uniformly sub-Gaussian then may $f(n)$ be chosen on the order of n ?

Note that we multiply the right side of (1) by the normalizing value $m^{-1/2}$ because if $x \in L_2([0, 1])$ then $\|x\|_{L_2([0,1])} = m^{-1/2} \mathbb{E}(\sum_{j=1}^m |x(t_j)|^2)^{1/2}$.

The first result on uniform stability for phase retrieval in an n -dimensional Hilbert space was for sampling m iid random vectors with uniform distribution on the sphere where m was on the order of $n \log(n)$ [CSV13], which was later improved to m being on the order of n [CL14]. This achievement was then extended to some other sub-Gaussian distributions [EM13] and to phase retrieval for μ -flat vectors using any sub-Gaussian distribution [KL18].

3. PHASE RETRIEVAL IN BANACH LATTICES AND CONTINUOUS SCHAUDER FRAMES

Our approach to phase retrieval for subspaces of $L_2(\mathbb{R})$ naturally generalizes to the Banach lattice setting. Let Z be a Banach lattice. We say that a subspace $X \subseteq Z$ *does stable phase retrieval* if there exists a constant $C > 0$ such that for all $f, g \in X$ we have that $\min(\|f - g\|, \|f + g\|) \leq C\||f| - |g|\|$. This leads to the following general questions.

Problem 3.1. Given a Banach lattice Z , how can we construct infinite dimensional subspaces $X \subseteq Z$ where phase retrieval is stable for X ? Can we classify up to isomorphism the infinite dimensional subspaces of Z where phase retrieval is stable? In particular, given $1 \leq p < \infty$, can we classify up to isomorphism the infinite dimensional subspaces of $L_p(\mathbb{R})$ where phase retrieval is stable?

For the case $2 \leq p < \infty$, it follows from classical Banach lattice techniques that if $X \subseteq L_p(\mathbb{R})$ does stable phase retrieval then X is isomorphic to a Hilbert space. However, Problem 3.1 is open for $1 \leq p < 2$. In the case of the Banach lattices $C([0, 1])$ and ℓ_∞ , we have checked that Banach's original proofs that every separable Banach space is isometric to subspaces of $C([0, 1])$ and ℓ_∞ actually construct subspaces where phase retrieval is stable.

In [AG17], the authors study phase retrieval for infinite dimensional spaces using continuous frames. If $(\phi_t)_{t \in \mathbb{R}}$ is a continuous frame of a Hilbert space H , then the analysis operator $\Phi : H \rightarrow L_2(\mathbb{R})$ given by $(\Phi(f))(t) = \langle f, \phi_t \rangle$ for all $t \in \mathbb{R}$ is an isomorphic embedding of H into $L_2(\mathbb{R})$. Thus phase retrieval for a continuous frame $(\phi_t)_{t \in \mathbb{R}}$ of H is equivalent to phase retrieval for a subspace $\Phi(H) \subseteq L_2(\mathbb{R})$. Like in the case for discrete frames, phase retrieval using a continuous frame for an infinite dimensional Hilbert space is always unstable [AG17]. Note that this does not contradict our results on stable phase retrieval for infinite dimensional subspaces of $L_2(\mathbb{R})$ as unlike in the discrete case, there exist subspaces of $L_2(\mathbb{R})$ which are not contained in the range of the analysis operator of a continuous frame. The main theorem in [AG17] goes even further, and they prove that phase retrieval using continuous Banach frames is unstable for every infinite dimensional Banach space.

There are many different generalizations of frames and continuous frames to the Banach space setting such as atomic decompositions, Banach frames, and Schauder frames. Atomic decompositions and Banach frames incorporate the analysis and synthesis operators for a frame into their definition. By composing the analysis and synthesis operators, one has a reconstruction formula for atomic decompositions and Banach frames. The definition of Schauder frame is more general as it is only defined in terms of the reconstruction formula for a frame. Recall that if X is a Banach space then $(x_j, f_j)_{j=1}^\infty \subseteq X \times X^*$ is called a *Schauder frame* for X if for all $x \in X$ we have that $x = \sum_{j=1}^\infty f_j(x)x_j$. The analysis and synthesis operators are fundamental to frame theory, and Schauder frames incorporate these operators through associated spaces. If $(x_j)_{j=1}^\infty$ is a frame for a Hilbert space H then the analysis operator embeds H into ℓ_2 . However, if $(x_j, f_j)_{j=1}^\infty$ is a Schauder frame for a Banach space X then it is not clear what Banach space the analysis operator should embed X into. If $(x_j, f_j)_{j=1}^\infty$ is a Schauder frame of X then a Banach space Z is called an *associated space* with *associated Schauder basis* $(z_j)_{j=1}^\infty$ if the map $x \mapsto \sum_{j=1}^\infty f_j(x)z_j$ is an embedding of X into Z and the map $\sum_{j=1}^\infty a_j z_j \mapsto \sum_{j=1}^\infty a_j x_j$ is a quotient map. One advantage of Schauder frames is that we can consider different associated spaces for a single Schauder frame $(x_j, f_j)_{j=1}^\infty$. Inspired by the results in [AG17], we propose extending the idea of associated spaces to continuous Schauder frames and considering phase retrieval in this new context.

Associated spaces for Schauder frames come with an associated basis, but it has been proven in many different ways that there is no such thing as a continuous basis over a measure space which isn't atomic (see [SB20] for example). We propose to define an associated space for a continuous Schauder frame in terms of Banach function spaces. If (M, Σ, μ) is a complete σ -finite measure space then a Banach space $X \subseteq L_0(\mu)$ is called a Banach function space on (M, Σ, μ) if there exists $u \in X$ with $u > 0$ a.e. and X satisfies the ideal property:

$$\text{For all } x \in L_0(\mu), \text{ if } \|x\| \leq |y| \text{ a.e. for some } y \in X \text{ then } x \in X \text{ and } \|x\| \leq \|y\|.$$

Let (M, Σ, μ) be a measure space and X be a separable Banach space. Suppose that $t \mapsto x_t \in X$ is w -measurable and $t \mapsto f_t \in X^*$ is w^* -measurable, then the pair $(x_t, f_t)_{t \in M}$ is called a *continuous Schauder frame of X over (M, Σ, μ)* if $x = \int_M f_t(x)x_t d\mu$ for all $x \in X$. Here the integral is defined weakly by the Pettis integral. We now define associated spaces for continuous Schauder frames. Let $(x_t, f_t)_{t \in M}$ be a

continuous Schauder frame for a Banach space X and let Z be a Banach function space over M . We call Z an *associated space* to $(x_t, f_t)_{t \in M}$ if the operators $T : X \rightarrow Z$ and $S : Z \rightarrow X$ are bounded, where $(Tx)(t) = f_t(x)$ for all $x \in X$ and $t \in M$, and $S(z) = \int z(t)x_t d\mu$ for all $z \in Z$. We call $T : X \rightarrow Z$ the analysis operator and $S : Z \rightarrow X$ the synthesis operator. Note that if μ is counting measure and the simple functions are dense in $Z \subseteq L_0(M)$ then the above definition reduces exactly to the case that $(x_t, f_t)_{t \in M}$ is an unconditional discrete Schauder frame and Z is an associated space with an unconditional associated basis. We propose extending the fundamental results about associated spaces for Schauder frames to the continuous setting. In particular, every continuous Schauder frame over a complete σ -finite measure space has an associated Banach function space. Indeed, if $(x_t, f_t)_{t \in M}$ is a continuous Schauder frame for X then we let $Z = \{z \in L_0(M) : \int z(t)x_t d\mu \in X\}$ and for $z \in Z$ we define $\|z\| = \sup_{E \in \Sigma} \|\int_E z(t)x_t d\mu\|$. We claim that if $x_t \neq 0$ a.e then this constructs an associated space Z of X . We now consider the question of using associated spaces for phase retrieval.

Problem 3.2. Let (M, Σ, μ) be a complete σ -finite measure space and let $(x_t, f_t)_{t \in M}$ be a continuous Schauder frame for a Banach space X . Suppose that Z is an associated space of $(x_t, f_t)_{t \in M}$ and that $T : X \rightarrow Z$ is the analysis operator. Phase retrieval for $T(X) \subseteq Z$ is stable if there exists $C > 0$ such that

$$\|x - y\|_X \leq C \| |T(x)| - |T(y)| \| \quad \text{for all } x, y \in X.$$

Is it possible for phase retrieval to be stable for $T(X) \subseteq Z$? What are necessary properties for Z to have so that phase retrieval may be stable for $T(X) \subseteq Z$? What are necessary properties for (M, Σ, μ) , $(x_t, f_t)_{t \in M}$, and X to have so that there exists an associated space Z such that phase retrieval to be stable for $T(X) \subseteq Z$? What are sufficient properties for (M, Σ, μ) , $(x_t, f_t)_{t \in M}$, and X to have so that there exists an associated space Z such that phase retrieval to be stable for $T(X) \subseteq Z$?

Note that ℓ_∞ is a Banach function space over $\mathbb{N}$, and that every Banach space with a 1-unconditional basis may be considered as a Banach function space over $\mathbb{N}$. We conjecture that every Banach space X with an unconditional basis has a continuous Schauder frame $(x_{(n,m)}, f_{(n,m)})_{(n,m) \in \mathbb{N} \times \mathbb{N}}$ with associated space $X \oplus \ell_\infty$ such that $T(X) \subseteq X \oplus \ell_\infty$ does stable phase retrieval. Thus we anticipate that our more general setting allows for stable phase retrieval in infinite dimensions using continuous frames. We conjecture that if phase retrieval is stable for $T(X) \subseteq Z$ then Z must be non-reflexive. Even more, we conjecture that if phase retrieval is stable for $T(X) \subseteq Z$ then Z must contain ℓ_∞ as a subspace.

The idea of an associated space for a continuous Schauder frame gives a useful way to extend phase retrieval using frames to the Banach space setting. However, it also opens up a lot of interesting intrinsic questions about continuous Schauder frames. Generally speaking, if we have a continuous Schauder frame for a Banach space with some important structure then we would like to construct an associated space which shares that structure. We are interested in the following questions in particular.

Problem 3.3. Let $(x_t, f_t)_{t \in M}$ be a continuous Schauder frame for a Banach space X . When does $(x_t, f_t)_{t \in M}$ have a reflexive associated Banach function space? If Y is a Banach space which is not isomorphic to a subspace of X , when does $(x_t, f_t)_{t \in M}$ have an associated Banach function space Z with the property that Y does not embed into Z ?

A discrete Schauder frame is shrinking if and only if it has an associated basis, and a Schauder frame is shrinking and boundedly complete if and only if it has a reflexive associated space [BFL12]. The properties shrinking and boundedly complete were extended to continuous Schauder frames in [EF20], and it is proven that a continuous Schauder frame for a separable Banach space X is shrinking if and only if ℓ_1 does not embed into X . Furthermore, a continuous Schauder frame for a separable Banach space X is shrinking and boundedly complete is equivalent to ℓ_1 and c_0 do not embed into X and also is equivalent to X being reflexive [EF20]. We conjecture that every continuous Schauder frame for a reflexive Banach space has a reflexive associated Banach function space. Furthermore, we conjecture that every continuous Schauder frame for a Banach space which does not contain ℓ_1 has an associated Banach function space that does not contain ℓ_1 .

There are many fundamental lattice properties that are useful in studying Banach function spaces. For example, a Banach function space Z is called *rearrangement invariant* if for all $x, y \in Z$ we have that $\|x\| = \|y\|$ whenever x and y have the same distribution. Z is called *order continuous* if whenever $(z_n)_{n=1}^\infty$ is a positive sequence in Z and $z_n \searrow 0$ we have that $\|z_n\| \rightarrow 0$. Z is said to have the *Fatou property* if

whenever $(z_n)_{n=1}^\infty$ is a norm bounded positive sequence in Z such that $0 \leq z_n \nearrow z \in L_0$ then $f \in Z$ and $\|z_n\| \rightarrow \|z\|$. There is a large established literature for Banach function spaces with these properties, and we can exploit this to study continuous Schauder frames.

Problem 3.4. Let $(x_t, f_t)_{t \in M}$ be a continuous Schauder frame for a Banach space X . When does $(x_t, f_t)_{t \in M}$ have an associated rearrangement invariant Banach function space? When does $(x_t, f_t)_{t \in M}$ have an associated order continuous Banach function space? When does $(x_t, f_t)_{t \in M}$ have an associated Banach function space with the Fatou property?

One of the useful ways to construct frames for Hilbert spaces and Schauder frames for Banach spaces is to embed them into a larger space and then project a basis onto the subspace. The orthogonal projection of an orthonormal basis for a Hilbert space is a Parseval frame for the subspace, and the projection of a Schauder basis for a Banach space is a Schauder frame for a subspace. Knowing when this technique will work in the continuous setting will give us a useful method for constructing continuous Schauder frames. We now consider the Hilbert space case for motivation. Let (M, Σ, μ) be a measure space and let $H \subseteq L_2(M)$ be a closed subspace. If H is a reproducing kernel Hilbert space over M then H is the range of the analysis operator of a continuous frame over (M, Σ, μ) . Indeed, if H is a RKHS over M then for all $t \in M$ there exists $f_t \in H$ such that $g(t) = \langle g, f_t \rangle$ for all $t \in M$. We have that $\|g\|^2 = \int_M |g(t)|^2 d\mu(t) = \int_M |\langle g, f_t \rangle|^2 d\mu(t)$ for all $g \in H$ and hence $(f_t)_{t \in M}$ is a continuous frame of H . The inclusion map of H into $L_2(M)$ is then the analysis operator of the frame $(f_t)_{t \in M}$ and orthogonal projection of $L_2(M)$ onto H is the synthesis operator. The following problem asks how this could be extended to Banach spaces.

Problem 3.5. Let Z be a Banach function space over (M, Σ, μ) and let $X \subseteq Z$ be a complemented subspace with a projection $P : Z \rightarrow X$. When is there an induced continuous frame $(x_t, f_t)_{t \in M}$ for X such that the analysis operator is the inclusion map of X into Z and the synthesis operator is the projection $P : Z \rightarrow X$?

In order to obtain $f_t \in X^*$ to solve Problem 3.5, we would need that point evaluation at t is a continuous linear functional on $X \subseteq Z$ and hence there exists $f_t \in X^*$ such that $f_t(x) = x(t)$ for all $x \in X$. We only need $(f_t)_{t \in M}$ for a continuous frame of a Hilbert space, but in the Banach space setting we also need $(x_t)_{t \in M}$. Obtaining $x_t \in X$ is more difficult and requires imposing additional structure. One approach would be to work with the idea of a reproducing kernel Banach space, which has been considered in the context of machine learning [MP04] [ZHZ09][ZZ17]. This would allow us to construct continuous Schauder frames which could be used for kernel methods in machine learning. Another approach would be to consider situations where for all $t \in M$ there is a natural net $(z_\alpha)_{\alpha \in E}$ in Z which converges in some sense to the distribution δ_t on M (where for all $s \in M$, $\delta_t(s) = 1$ if and only if $s = t$). We could then consider properties of $X \subseteq Z$ where the limit $\lim_{\alpha \in E} Pz_\alpha$ exists and let $x_t = \lim_{\alpha \in E} Pz_\alpha$. We anticipate that a development of both approaches will lead to new useful methods of constructing continuous Schauder frames.

4. SAMPLING OF CONTINUOUS FRAMES

Recall that if H is a separable Hilbert space, then a collection of vectors $(x_j)_{j \in J} \subset H$ is called a *frame* for H if there exists constants $0 < A \leq B < \infty$ such that $A\|x\|^2 \leq \sum \langle x, x_j \rangle^2 \leq B\|x\|^2$ for all $x \in H$. A continuous frame generalizes this by replacing the series with an integral, and the following definition formalizes this notion. Let (X, μ) be a positive, σ -finite measure space and let H be a separable Hilbert space. A measurable function $\Psi : X \rightarrow H$ is a *continuous frame* of H with respect to μ if there exists constants $A, B > 0$ such that $A\|x\|^2 \leq \int |\langle x, \Psi(t) \rangle|^2 d\mu(t) \leq B\|x\|^2$ for all $x \in H$. The constant A is called a *lower frame bound* and the constant B is called an *upper frame bound*. If $A = B$ then the continuous frame is called *tight* and if $A = B = 1$ then the continuous frame is called *Parseval*.

Continuous frames are widely used in mathematical physics and are particularly prominent in quantum mechanics and quantum optics. Though continuous frames such as the short time Fourier transform naturally characterize many different physical properties, discrete frames are much better suited for computations. Because of this, when working with continuous frames, researchers often create a discrete frame by sampling the continuous frame and then use the discrete frame for computations instead of the entire continuous frame. Specifically, if $\Psi : X \rightarrow H$ is a continuous frame and $(t_j)_{j \in J} \subseteq X$ then $(\Psi(t_j))_{j \in J} \subseteq H$ is called a *sampling* of Ψ . The discretization problem, posed by Ali, Antoine, and Gazeau in their physics textbook *Coherent States, Wavelets, and Their Generalizations* [AAG00], asks when a continuous frame of a Hilbert

space can be sampled to obtain a frame. They state that a positive answer to the question is crucial for practical applications of coherent states, and chapter 16 of the book is devoted to the discretization problem. The PI with Darrin Speegle solved the discretization problem in its full generality by characterizing exactly when a continuous frame may be sampled to obtain a frame [FS19]. Our result can be quantified as, there exists $C, D > 0$ such that if $\Psi : X \rightarrow H$ is a continuous Parseval frame with $\|\Psi(t)\| \leq 1$ for all $t \in X$ then there exists $(t_j)_{j \in J} \in X^J$ such that $(\Psi(t_j))_{j \in J}$ is a frame of H with lower frame bound C and upper frame bound D .

It is important for applications to know that a given continuous frame can be sampled to obtain a frame. However, not all frames are equally well suited for applications, and often additional properties are desired for a frame. Localization is a particularly important concept in applications and was developed independently by Gröchenig [G91] and Balan, Casazza, Heil, and Landau [BCHL06]. If $(f_j)_{j \in J}$ and $(e_k)_{k \in G}$ are two frames for a Hilbert space H we can consider the relationship between how vectors are represented in the two coordinate systems. We can think of $(f_j)_{j \in J}$ as being localized with respect to $(e_k)_{k \in G}$ if when we consider a vector $x \in H$ whose representation in $(e_k)_{k \in G}$ is distributed tightly around a small number of coordinates then the representation in $(f_j)_{j \in J}$ will be distributed tightly around some corresponding coordinates. Formalizing this concept can vary based on the circumstances and here we follow the approach initiated in [BCHL06]. Let $(f_j)_{j \in J}$ and $(e_k)_{k \in G}$ be frames for a Hilbert space H where G is a discrete countable abelian group and let $a : I \rightarrow G$ be a map. We say that $((f_j)_{j \in J}, a, (e_k)_{k \in G})$ is ℓ_1 localized if

$$\sum_{k \in G} \sup_{j \in J} |\langle f_j, e_{a(j)-k} \rangle| < \infty.$$

We say that $(e_k)_{k \in G}$ is ℓ_1 self localized if $((e_k)_{k \in G}, Id_G, (e_k)_{k \in G})$ is ℓ_1 localized.

Using the theory of coorbit spaces, Fornasier and Rauhut proved that certain classes of continuous frames may be sampled to obtain discrete frames [FoR05]. Their method has the further benefit in that if the given continuous frame is self localized then the resulting sampled frame will be self localized as well. This leads to the following questions.

Problem 4.1. Given a Hilbert space H , when can a continuous frame of H be sampled to obtain a self-localized frame of H ? Given a Hilbert space H and a frame $(e_k)_{k \in G}$ of H , when can a continuous frame of H be sampled to obtain a frame which may be localized with respect to $(e_k)_{k \in G}$?

Our solution to the discretization problem does not preserve any localization properties when sampling. The main reason for this is that our proof relies heavily on the following theorem of Marcus, Spielman, and Srivastava.

Theorem 4.2 (MSS15 Cor 1.5). *Let $(x_j)_{j \in J} \subseteq H$ be a sequence with Bessel bound 1 and $\|x_j\|^2 \leq \delta$ for all $j \in J$. Then there exists a partition $\{J_1, J_2\}$ of J such that both $(x_j)_{j \in J_1}$ and $(x_j)_{j \in J_2}$ have Bessel bound $(1/2 + (2\delta)^{1/2} + \delta)$.*

The above theorem solves the Kadison-Singer Problem and allows frames to be partitioned in a way that uniformly reduces the upper frame bound. However, this does not uniformly reduce the localization constant. The Kadison-Singer Problem had been shown to be equivalent to many famous problems in a variety of areas, including the Feichtinger Conjecture which states that every frame of unit vectors is a finite union of Riesz sequences. Notably, the Feichtinger Conjecture had already been proven for the case of ℓ_1 self localized frames [BCHL06]. Our goal is to incorporate the localization properties into the interlacing families technique in [MSS15] in order to both solve Problem 4.1 and to obtain better constants for the Feichtinger conjecture in the case of localized frames. This could also lead to stronger applications of the interlacing families technique in other areas as well.

Let (M, μ) be a positive, σ -finite measure space and let H be a separable Hilbert space. Let $t \mapsto \psi_t$ and $t \mapsto \phi_t$ be two measurable maps from M to H . Then $(\psi_t, \phi_t)_{t \in M}$ is called a *reproducing pair* if the map $g \oplus f \mapsto \int \langle g, \psi_t \rangle \langle f, \phi_t \rangle d\mu$ defines a bounded and invertible operator on $H \oplus H$. Continuous frames are reproducing pairs for the case $\psi_t = \phi_t$ for all $t \in M$. Many of the important properties of continuous frames extend to reproducing pairs, but reproducing pairs allow for more flexibility in their construction and do not require each map individually to satisfy frame bounds [AST17][SB15][SB17]. This leads to the following question on sampling reproducing pairs.

Problem 4.3. Let $(\psi_t, \phi_t)_{t \in M}$ be a reproducing pair for separable Hilbert space H . When does there exist a countable subset $J \subseteq M$ such that $x \mapsto \sum_{j \in J} \langle x, \psi_j \rangle \phi_j$ defines a bounded and invertible operator on H and the series converges unconditionally?

Because of the use of Theorem 4.2 in solving the discretization problem, any progress on Problem 4.2 has strong potential to lead to a solution for Problem 4.3 in certain cases.

Instead of considering when a continuous frame can be sampled to obtain a discrete frame which preserves the property of being localized, we now consider the problem of sampling continuous frames to obtain a discrete frame which does stable phase retrieval. As continuous frames never do stable phase retrieval for infinite dimensional Hilbert spaces, we are necessarily considering continuous frames for finite dimensional Hilbert spaces. Let $\Psi : M \rightarrow H$ be a continuous Parseval frame over a measure space (M, μ) . We say that Ψ does C -stable phase retrieval for some constant $C > 0$ if whenever $x, y \in H$ we have that $\min(\|x - y\|, \|x + y\|) \leq C \|\langle x, \Psi(t) \rangle - \langle y, \Psi(t) \rangle\|_{L_2(\mu)}$.

Problem 4.4. Let $C > 0$. Do there exist universal constants $A, B, D > 0$ such that if $\Psi : X \rightarrow H$ is a continuous Parseval frame with $\|\Psi(t)\| \leq 1$ for all $t \in X$ which does C -stable phase retrieval then there exists $(t_j)_{j \in J} \in X^J$ such that $(\Psi(t_j))_{j \in J}$ is a frame of H with lower frame bound A and upper frame bound B which does D -stable phase retrieval?

In Section 2 we considered obtaining frames for $\mathbb{R}^n$ which do stable phase retrieval by choosing k independent random vectors with some prescribed distribution. Under a sub-Gaussian assumption on the distribution we had with high probability that the k -random vectors did stable phase retrieval with stability constant independent of the dimension n with k chosen on the order of n . A random vector with expectation $\mathbb{E} = Id_{\mathbb{R}^n}$ is simply a continuous Parseval frame over a probability space. Thus, by making sub-Gaussian assumptions, with high probability we can choose suitable sampling points in Problem 4.4 by choosing $(t_j)_{j=1}^k$ at random. However, without the sub-Gaussian assumption we could only obtain uniform stability with k chosen on the order of $n \log(n)$ and hence the frame bound of the sampled frame would be on the order of at least $n \log(n)$. Hence, in solving Problem 4.4 our goal is to prove that such a sampling is possible even though it may happen with very small probability.

5. CONSTRUCTING COORDINATE SYSTEMS FOR $L_p(\mathbb{R})$

This section is concerned with developing techniques to build coordinate systems with new properties in the classical Banach spaces $L_p(\mathbb{R})$ for $1 < p < \infty$. Though we do not discuss phase retrieval in this section, the mathematical ideas we are proposing fit in well with the other sections of the proposal. We will be focusing on creating coordinate systems of translations of a single function and coordinate systems of positive functions. We will first discuss the structure of translations of a single function before discussing positive functions and explaining some nice connections between these two seemingly unrelated problems.

Given $\lambda \in \mathbb{R}$ and a function $f \in L_2(\mathbb{R})$, $T_\lambda(f)$ is the translation of f by λ . That is, $T_\lambda(f)(x) = f(x - \lambda)$ for all $x \in \mathbb{R}$. The fundamental role that the translations of a single function play in both wavelet theory and Gabor analysis has motivated the study of coordinate systems formed by sets of the form $\{T_\lambda f\}_{\lambda \in \Lambda}$ and the study of subspaces of the form $\overline{\text{span}}\{T_\lambda f\}_{\lambda \in \Lambda}$. A simple Fourier transform argument yields that there is no function $f \in L_2(\mathbb{R})$ such that $\overline{\text{span}}\{T_n f\}_{n \in \mathbb{Z}} = L_2(\mathbb{R})$. It is then very surprising that if $\{\lambda_n\}_{n \in \mathbb{Z}} \subset \mathbb{R} \setminus \mathbb{Z}$ is such that $\lim_{n \rightarrow \pm\infty} |\lambda_n - n| = 0$, then there exists $f \in L_2(\mathbb{R})$ such that $\overline{\text{span}}\{T_{\lambda_n} f\}_{n \in \mathbb{Z}} = L_2(\mathbb{R})$ [O79]. However, there does not exist a set $\Lambda \subset \mathbb{R}$ such that $\{T_\lambda f\}_{\lambda \in \Lambda}$ is a Hilbert frame for $L_2(\mathbb{R})$ [CDH99]. Given a collection of vectors $(x_j)_{j \in J}$ in a Hilbert space, it is possible that $(x_j)_{j \in J}$ is not a frame but that there exists scalars $(\alpha_j)_{j \in J}$ such that $(\alpha_j x_j)_{j \in J}$ is a frame. This idea of scaling a frame leads to many interesting questions [BAG10],[SB13],[KOPT13], and we introduce this idea to the situation of translations of a single function.

Problem 5.1. Does there exist a function $f \in L_2$, a set $\Lambda \subset \mathbb{R}$, and scalars $\{\alpha_\lambda\}_{\lambda \in \Lambda}$ such that $\{\alpha_\lambda T_\lambda f\}_{\lambda \in \Lambda}$ is a frame for L_2 ?

It is necessary for Λ to be uniformly discrete in order for the set of translations $\{T_\lambda f\}_{\lambda \in \Lambda}$ to be Bessel, but if the scalars $\{\alpha_\lambda\}_{\lambda \in \Lambda}$ are not bounded below then it is possible for the scaled translations $\{\alpha_\lambda T_\lambda f\}_{\lambda \in \Lambda}$ to be Bessel and have Λ not be uniformly discrete. This distinction adds a subtle difficulty in answering Problem 5.1.

It is known that there exists $f \in L_p(\mathbb{R})$ such that $\overline{\text{span}}\{T_n f\}_{n \in \mathbb{Z}} = L_p(\mathbb{R})$ if and only if $2 < p < \infty$ [AO96]. The PI, E. Odell, Th. Schlumprecht, and A. Zsák strengthened that result by proving that there exists a function $f \in L_p(\mathbb{R})$ and sequence $(g_n^*)_{n \in \mathbb{N}} \subset L_p^*(\mathbb{R})$ such that $(T_n f, g_n^*)_{n \in \mathbb{N}}$ forms an unconditional Schauder frame for $L_p(\mathbb{R})$ if (and only if) $2 < p < \infty$ [FOSZ14]. Thus, although a frame of translates of a single function is impossible for a Hilbert space, it is possible in the more general Banach space setting.

We now define Schauder frames and Schauder bases for Banach spaces. Let X be a Banach space. A sequence of vectors $(x_j)_{j=1}^\infty$ is called a *Schauder basis* for X if for all $x \in X$ there exists a unique sequence of scalars $(a_j)_{j=1}^\infty$ such that $x = \sum_{j=1}^\infty a_j x_j$. A Schauder frame is both a generalization of a Schauder basis for a Banach space and a generalization of dual frames for Hilbert spaces. Let X be a Banach space with dual X^* . A sequence of pairs of vectors and functionals $(x_j, f_j)_{j=1}^\infty \subseteq X \times X^*$ is called a *Schauder frame* for X if for all $x \in X$ we have that $x = \sum_{j=1}^\infty f_j(x) x_j$. We know that $L_p(\mathbb{R})$ has an unconditional Schauder frame of translates of a single function for $p > 2$, and the PI has recently proven with A. Powell and M. Taylor that $L_p(\mathbb{R})$ has a Schauder frame of translates of a single function for all $1 \leq p < \infty$ [FPT20]. It is known that $L_1(\mathbb{R})$ cannot have a Schauder basis of translates of a single function. The question for Schauder bases is still open for all $1 < p < \infty$ though.

Problem 5.2. Does there exist $1 < p < \infty$, $f \in L_p(\mathbb{R})$, and $(\lambda_n)_{n \in \mathbb{N}} \subset \mathbb{R}$ such that $(T_{\lambda_n} f)_{n \in \mathbb{N}}$ is a Schauder basis for $L_p(\mathbb{R})$?

We now consider the problem of constructing coordinate systems of positive functions for $L_p(\mathbb{R})$. The admisability condition for wavelets states that if a function $\psi \in L_2(\mathbb{R})$ is a wavelet, then it must satisfy $\int \psi(t) dt = 0$. In particular, wavelets cannot be positive vectors. This naturally leads to the question, what kind of coordinate systems can be constructed from positive vectors? A. Powell and A. Spaeth considered this general problem and proved that for all $1 \leq p < \infty$, there do exist positive Schauder frames and Markeschevitz bases for $L_p(\mathbb{R})$ but there do not exist positive unconditional Schauder frames or positive unconditional Schauder bases for $L_p(\mathbb{R})$ [PS16]. They listed multiple open questions on the existence of positive coordinate systems, the most significant being, given $1 \leq p < \infty$, does there exist a positive Schauder basis for $L_p(\mathbb{R})$? W.B. Johnson and G. Schechtman solved this problem for the case $p = 1$ by proving that $L_1(\mathbb{R})$ does have a positive Schauder basis [JS15]. This leaves the following remaining problem.

Problem 5.3. For each $1 < p < \infty$, does there exist a positive Schauder basis for $L_p(\mathbb{R})$?

In [PS16], it is proven that any positive Schauder basis for $L_p(\mathbb{R})$ must be conditional and cannot have basis constant 1. Our goal is to construct for all $\varepsilon > 0$ and all $1 < p < \infty$ a positive conditional Schauder basis for $L_p(\mathbb{R})$ with basis constant $(1 + \varepsilon)$. This would give the best possible solution to Problem 5.3. All of the previously constructed conditional bases for $L_p(\mathbb{R})$ are of course not positive, but for all $1 < p < \infty$ and $\varepsilon > 0$ there do exist conditional positive $(1 + \varepsilon)$ -basic sequences in $L_p(\mathbb{R})$. The idea for the Johnson-Schechtman construction is to break up a vector f into a positive part f^+ and a negative part f^- then append a vector with disjoint support and equal norm to both parts. This works in $L_1(\mathbb{R})$ but fails for $L_p(\mathbb{R})$ for all $1 < p < \infty$. Our idea is to break up the positive part of f into N pieces with equal distribution and the negative part of f into N pieces with equal distribution. We then construct a positive highly conditional basic sequence $(x_i)_{i=1}^{2N}$ with disjoint support from f and append the negative pieces onto $(x_{2n-1})_{n=1}^N$ and the positive pieces onto $(x_{2n})_{n=1}^N$. Given $\varepsilon > 0$, the conditionality allows for $f = f^+ - f^-$ to be approximated within ε , but for the closest linear combination to $f^+ + f^-$ to have norm smaller than ε . We propose formalizing and expanding upon this idea to solve Problem 5.3.

Coincidentally, our idea for constructing positive conditional basic sequences that we are working on to solve Problem 5.3 can be slightly modified to give a new construction for a conditional basic sequence of translations of a single function. Thus the techniques we are building to solve Problem 5.3 can also be used to attack Problem 5.2. In order to have a Schauder basis $(x_j)_{j=1}^\infty$ for $L_p(\mathbb{R})$, the initial segments of $(x_j)_{j=1}^\infty$ must be uniformly complemented in $L_p(\mathbb{R})$. This is not a problem for $p = 2$, but it is a significant obstacle for $p \neq 2$. In order to avoid that requirement we could ask for a total minimal system instead of a Schauder basis. A sequence $(x_j)_{j=1}^\infty$ is called a total minimal system in a Banach space X if $(x_j)_{j=1}^\infty$ has dense span in X and $(x_j)_{j \neq n}$ does not have dense span in X for all $n \in \mathbb{N}$. It is known that $L_p(\mathbb{R})$ has a total minimal system of positive functions for all $1 < p < \infty$ [PS16] and that $L_p(\mathbb{R})$ has a total minimal system of translations of a single function for all $2 \leq p < \infty$. Furthermore, $L_1(\mathbb{R})$ does not have a total minimal system of translations of a single function. This leaves the following problem open.

Problem 5.4. For each $1 < p < 2$, does there exist a total minimal system for $L_p(\mathbb{R})$ consisting of translations of a single function?

Let X be a separable Banach function space X over $\mathbb{R}$. Olevskii and Ulanovskii recently developed a technique for constructing a Schwartz function $f \in X$ whose translates by small perturbations of integers are dense in X [OU18]. We propose combining our techniques with those in [OU18] to attack Problem 5.4.

6. FRAMES FOR VECTOR BUNDLES

Previous topics have discussed frames for Hilbert spaces and Schauder frames for Banach spaces. In differential topology and geometry the word frame has a different meaning. In this context, if M is a smooth manifold, then a moving frame for M is a smoothly varying basis for the tangent space at each point in M . To avoid confusion, we will refer to these as moving bases. Though moving bases are important and useful, they do not always exist. For example, the 2-sphere does not have a moving basis or even a nowhere zero vector field. One way to overcome this problem is to combine the two notions of frames. If M is a smooth manifold and $V = \cup_{x \in M} V_x$ is a vector bundle over M then a *section* of V is a continuous function $f : M \rightarrow V$ such that $f(x) \in V_x$ for all $x \in M$.

If M is a smooth manifold and $V = \cup_{x \in M} V_x$ is a vector bundle over M , then we say that a sequence of sections $(f_i)_{i=1}^k$ of V is a *moving redundant frame* or simply *moving frame* for V if the sequence $(f_i(x))_{i=1}^k$ is a frame for the vector space V_x for all $x \in M$. We say that $(f_i)_{i=1}^k$ is a *moving Parseval frame* for V if $(f_i(x))_{i=1}^k$ is a Parseval frame for the vector space V_x for all $x \in M$. We say that $(f_i)_{i=1}^k$ is a *moving finite unit norm tight frame (FUNTF)* for V if $(f_i(x))_{i=1}^k$ is a tight frame of unit vectors for the vector space V_x for all $x \in M$.

As frames are defined in terms of an inner product, we are necessarily considering vector bundles with a given moving inner product. Fortunately, a standard partition of unit argument yields that every vector bundle over a paracompact manifold may be endowed with a moving inner product. In the case that we are considering the tangent bundle of a Riemannian manifold, then we take the Riemannian metric to be the moving inner product.

It immediately follows from the definition of moving Parseval frame that every n -dimensional smooth manifold has a moving Parseval frame of $(2n - 1)$ -vectors for its tangent bundle. Indeed, one can immerse the manifold into $\mathbb{R}^{2n-1}$ and then project a fixed basis on to the tangent space of the manifold at each point. Thus in sharp contrast to moving bases, we have that moving Parseval frames can always be found. An application of this concept was shown by P. Kutchment [K09] who proved that particular vector bundles over the torus which arise in mathematical physics have natural moving redundant frames of analytic functions but do not have moving bases of analytic functions.

Many of the interesting ideas and questions in this research area are accessible to strong undergraduate students. With this in mind, the principal investigator previously created a project for a Research Experience for Undergraduates with the intention of building some of the basic theory for moving redundant frames and creating interesting examples of moving redundant frames. In collaboration with three undergraduate students, the principle investigator extended the celebrated Naimark dialation theorem to the manifold setting by proving that if $(f_i)_{i=1}^k$ is a moving redundant frame for a smooth manifold M , then M embeds as a submanifold of a smooth manifold X such that X has a moving basis which projects to $(f_i)_{i=1}^k$ [FPWW14]. Later, the principal investigator worked with two other undergraduates to prove that the n -sphere S^n has a moving redundant finite unit norm tight frame for its tangent space if and only if n is odd [FHM14]. This is particularly interesting as the only spheres which have a moving basis for their tangent space are S^1 , S^3 , and S^7 . Eileen Marin, one of the undergraduates collaborators, expanded on this work for her undergraduate thesis under the principal investigator's guidance and was awarded a Goldwater Fellowship, the most prestigious undergraduate scholarship given in the natural sciences, engineering and mathematics. We propose to continue this successful undergraduate research program by combining phase retrieval with moving redundant frames on manifolds.

let M be a smooth manifold and $V = \cup_{x \in M} V_x$ be a vector bundle over M . We denote the space of sections of V by Γ . Note that Γ is an infinite dimensional Banach space when considered as the closed subspace of $C(M, V)$ which consists of all continuous functions $f : M \rightarrow V$ such that $f(x) \in V_x$ for all $x \in M$. In particular, for $f \in \Gamma$ we have that $\|f\|_\Gamma = \sup_{x \in M} \|f(x)\|_{V_x}$. We define an equivalence relation $\sim$ on Γ by $f \sim g$ if for all $x \in M$ there exists $\varepsilon_x = \pm 1$ such that $f(x) = \varepsilon_x g(x)$. We say that a moving redundant frame

$(f_j)_{j=1}^k$ for V does *phase retrieval* if for any two sections $f, g : \Gamma$ if $(|\langle f(x), f_j(x) \rangle|)_{j=1}^k = (|\langle g(x), f_j(x) \rangle|)_{j=1}^k$ for all $x \in M$ then $f \sim g$. Likewise, we say that $(f_j)_{j=1}^k$ does *stable phase retrieval* if there exists a constant C so that for any two sections $f, g \in \Gamma$,

$$(2) \quad \inf_{h \sim g} \|f - h\|_{\Gamma} \leq C \left(\sup_{x \in M} \sum_{j=1}^k \left| |\langle f, f_j \rangle| - |\langle g, f_j \rangle| \right|^2 \right)^{1/2}.$$

We now consider the general question of how do the general principles and theorems for phase retrieval extend to this continuous setting? One might expect that many of these results would pass through with minor adjustment to the continuous setting, but surprisingly that is not the case. For example, a frame for a finite dimensional Hilbert space does phase retrieval if and only if it does stable phase retrieval [CCPW13][B17]. This fundamental result can fail dramatically for moving frames for vector bundles. Indeed, if one considers the natural embedding $S^2 \subset \mathbb{R}^3$ then the projection of the unit vector basis of $\mathbb{R}^3$ onto the tangent space of S^2 gives a moving frame $(f_j)_{j=1}^3$ of 3 vector fields for TS^2 which does phase retrieval. However, one can use the hairy ball theorem to prove that there does not exist a moving frame of 3 vector fields for S^2 which does stable phase retrieval for TS^2 . The reason for this discrepancy is that we can use the continuity of a vector field when doing phase retrieval and thus it is possible for the moving frame $(f_j)_{j=1}^3$ to do phase retrieval for TS^2 but for it not to be the case that for each $x \in S^2$, the fixed frame $(f_j(x))_{j=1}^3$ does phase retrieval for $T_x S^2$. We give the following conjecture concerning characterizing when a moving frame does phase retrieval or stable phase retrieval for a vector bundle.

Conjecture 1. Let $(f_j)_{j=1}^k$ be a moving frame for a vector bundle V over a manifold M . Then $(f_j)_{j=1}^k$ does phase retrieval for V if and only if the set of points $x \in M$ where $(f_j(x))_{j=1}^k$ does phase retrieval for V_x is dense in M . Furthermore, $(f_j)_{j=1}^k$ does stable phase retrieval for V if and only if for all $x \in M$, $(f_j(x))_{j=1}^k$ does phase retrieval for V_x .

The Whitney Immersion Theorem gives that every n -dimensional real manifold immerses into $\mathbb{R}^{2n-1}$. Thus, every n -dimensional smooth manifold has a moving Parseval frame of $2n - 1$ vectors for its tangent bundle which can be constructed by immersing the smooth manifold M in $\mathbb{R}^{2n-1}$ and then projecting the basis for $\mathbb{R}^{2n-1}$ onto TM . Likewise, a generic frame of $2n - 1$ vectors in $\mathbb{R}^n$ does phase retrieval and no frame of less than $2n - 1$ vectors in $\mathbb{R}^n$ does phase retrieval. Thus, every smooth manifold has a moving Parseval frame for its tangent bundle which has the necessary number of vector fields which are required to do phase retrieval.

Problem 6.1. Does every n -dimensional smooth manifold have a moving Parseval frame of $2n - 1$ vector fields which does phase retrieval for the tangent bundle?

The previous problems we have considered deal with understanding when phase retrieval is possible. We now consider the problem of developing algorithms for implementing phase retrieval. One difficulty is that when trying to reconstruct a section f , any algorithm can only consider the values $(|\langle f(x), f_j(x) \rangle|)_{j=1}^k$ for finitely many $x \in M$. Thus our goal is to construct an algorithm which chooses points $(x_i)_{i=1}^n$ in M and then based on the values $(|\langle f(x_i), f_j(x_i) \rangle|)$ it chooses a section g which should be close in norm to a section h with $h \sim f$. As any implementation of an algorithm will include error, we desire to also have an estimate for the stability of the algorithm.

Problem 6.2. Let $(f_j)_{j=1}^k$ be a moving frame for a vector bundle V over a manifold M . How can we create an algorithm for doing phase retrieval for this moving redundant frame? Assuming some property of a section f of V (such as a Lipschitz bound) how could we bound the maximal error of the algorithm? Given a certain bound on the error in measurements $|\langle f(x), f_j(x) \rangle|$, such as the standard ℓ_2 bound, how could we bound the maximal error of the algorithm?

Broader Impacts

The PI and CO-PI have obtained approval from our department to develop a course in frame theory intended for graduate and undergraduate students to be taught in the spring semesters. We intend to supplement the course with research visitors during the semester as well as a summer research program for graduate and undergraduate students. Our goal for visitors is to collaborate on research with them and for them to give two talks while they are visiting: one traditional seminar talk and one directed to students in the frame theory course. We are particularly excited about the latter; being able to meet and learn from mathematicians from outside the university is a special opportunity for students. Our goal for the summer research program is to support highly motivated undergraduate and graduate students from the course to continue their study of frame theory and begin work on a research project with the PI and CO-PI. The aforementioned proposed project on phase retrieval for sections of vector bundles is designed to be collaborative and contains multiple avenues of research that are accessible for students. The PI has previously been very successful in mentoring undergraduate research. Past accomplishments include writing two papers with a total of five undergraduate coauthors [FPWW14][FHM14], and directing Eileen Martin's honors senior thesis [M12], who was awarded a Goldwater Scholarship for the project.

A student chapter of the Association for Women in Mathematics (AWM) has recently been created at St Louis University, and as a former chapter president, the Co-PI is uniquely poised to support and grow this new student group. Through the Co-PI's liasonship with the St. Louis University AWM student chapter, we are well-positioned to directly engage with the women mathematicians from this group, and for those that are interested, provide meaningful learning experiences through the proposed course, summer research program, mentorship, and opportunities to meet and have dinner with women mathematicians that visit as a part of this grant. The grant will also allow the Co-PI to bring the students to the Nebraska conference for undergraduate women in math where the students could present their research.

REFERENCES

- [AAG00] S.T. Ali, J.-P. Antoine, J.-P. Gazeau, *Coherent States, Wavelets, and Their Generalizations*. Graduate Texts in Contemporary Physics (Springer, New York, 2000)
- [AG17] R. Alaifari and P. Grohs, *Phase retrieval in the general setting of continuous frames for banach spaces*, SIAM Journal on Mathematical Analysis, **49** (3) (2017), 1895–1911.
- [AO96] A. Atzmon and A. Olevskii, *Completeness of integer translates in function spaces on $\mathbb{R}$* , J. Approx. Theory **87** (1996), 291– 327.
- [AST17] J.P. Antoine, M. Speckbacher, and C. Trapani, *Reproducing pairs of measurable functions*, Acta Applicandae Math., **150** (2017), no. 1, 81-101.
- [BAG10] P. Balazs, J.P. Antoine, and A. Grybos, *Weighted and controlled frames: Mutual relationship and first numerical properties*, Int. J. Wavelets Multiresolut. Inf. Process. **8** (1) (2010), 109–132.
- [B17] R. Balan, *Stability of frames which give phase retrieval*, Houston journal of mathematics, **43** (3) (2017), 905-918.
- [BCE06] R. Balan, P.G. Casazza, and D. Edidin, *On Signal Reconstruction Without Phase*, Appl. and Compt. Harmonic Analysis, **20** (3) (2006), 345-356.
- [BCHL06] R. Balan, P.G. Casazza, C. Heil, and Z. Landau, *Density, overcompleteness, and localization of frames I: Theory*, J. Fourier Anal. Appl., **12** (2006, no. 2, 105-143.
- [BW15] Balan, R. and Wang, Y., 2015. Invertibility and robustness of phaseless reconstruction. Applied and Computational Harmonic Analysis, 38(3), pp.469-488.
- [BFL12] K. Beanland, D. Freeman, and R. Liu, *Upper and lower estimates for Schauder frames and atomic decompositions*, Fund. Math. **231** (2015), 161-188.
- [BoHa] B. Bodmann and N. Hammen, *Stable Phase retrieval with low redundancy frames*, Adv Comput Math **41**, 317–331 (2015)
- [BCGJT16] Botello-Andrade, Sara, Peter G. Casazza, Dorsa Ghoreishi, Shani Jose, and Janet C. Tremain, *Weak phase retrieval and phaseless reconstruction*, arXiv preprint arXiv:1612.08018 (2016)
- [BR99] C. Becchetti and L. P. Ricotti. Speech recognition theory and C++ implementation. Wiley (1999).
- [CCPW16] J. Cahill, P.G. Casazza, J. Peterson, and L. Woodland, *Phase retrieval by projections*, Houston J. Math. **42**, No. 2 (2016), 537-558.
- [CCPW13] J. Cahill, P.G. Casazza, J. Peterson, and L. Woodland, *Using projections for phase retrieval*, Proc. SPIE **8858**, Wavelets and Sparsity XV, 88581W (26 September 2013)
- [CoEdHe15] A. Conca, D. Edidin, M. Hering, and C. Vinzant, *An algebraic characterization of injectivity of phase retrieval*, Appl. Comput. Harmon. Anal., **38** (2) (2015) 346-356.
- [CCD16] J. Cahill, P.G. Casazza, and I. Daubechies, *Phase retrieval in infinite dimensional Hilbert spaces*, Transactions of the AMS, Series B, **3** (2016), 63-76.
- [CDH99] O. Christensen, B. Deng, and C. Heil, *Density of Gabor frames*, Appl. Comput. Harmon. Anal. **7** (1999), 292 – 304.
- [CDFF20] R. Calderbank, I. Daubechies, D. Freeman, and N. Freeman, *Stable phase retrieval from magnitude of point evaluation for infinite dimensional subspaces of $L_2(\mathbb{R})$* , in preparation.
- [CSV13] E. J. Candès, T. Strohmer, and V. Voroninski *Phaselift: Exact and stable signal recovery from magnitude measurements via convex programming*, Comm. Pure Appl. Math., **66** (8) (2013), 1241-1274.
- [CL14] E.J. Candès, X. Li, *Solving quadratic equations via PhaseLift when there are about as many equations as unknowns*, Found Comput Math **14** (5) (2014), 1017-1026.

- [CGJT17] Casazza, Peter, Dorsa Ghereishi, Shani Jose, and Janet Tremain. "Norm retrieval and phase retrieval by projections." *Axioms* 6, no. 1 (2017): 6.
- [CE15] A. Conca, D. Edidin, M. Hering, and C. Vinzant, An algebraic characterization of injectivity of phase retrieval, *Appl. Comput. Harmon. Anal.*, 38 (2) (2015) 346-356.
- [Ed15] Edidin, D. Projections and Phase retrieval. *Appl. Comput. Harmonic Anal.* 2017, 42, 350–359.
- [EF20] J. Eisner, D. Freeman, *Continuous Schauder Frames for Banach Spaces*, *J Fourier Anal. Appl.*, **26**, 66 (2020).
- [EM13] Y. C. Eldar and S. Mendelson, *Phase retrieval: stability and recovery guarantees*, *Appl. Comput. Harmonic Analysis*, **36** (3), (2013) 473-494.
- [FHM14] D. Freeman, R. Hotovy, and E. Martin, *Moving finite unit norm tight frames for S^n* , *Illinois J. Math.*, **58** (2014), no. 2, 311-322.
- [FOSZ14] D. Freeman, E. Odell, Th. Schlumprecht, A. Zsak, *Unconditional structures of translates for $L_p(\mathbb{R}^d)$* , *Israel J. of Math.*, 203 (2014), no. 1, 189-209.
- [FPT20] D. Freeman, A. Powell, and M. Taylor, *A Schauder basis for L_2 consisting of non-negative functions*, preprint.
- [FPWW14] D. Freeman, D. Poore, R. Wei, M. Wyse, *Moving Parseval frames for vector bundles*, *Houston J. Math.*, **40** (2014), no 3, 817-832.
- [FS19] D. Freeman and D. Speegle, *The discretization problem for continuous frames*, *Advances in Math.*, **345** (2019), 784-813.
- [FoR05] M. Fornasier, H. Rauhut, *Continuous frames, function spaces, and the discretization problem*, *J. Fourier Anal. Appl.* **11** (2005), 245-287.
- [G91] K. Gröchenig, *Describing functions: Atomic decompositions versus frames*, *Monatsh. Math.*, **112**, 1 (1991), 1-42.
- [HLLL14] D. Han, D.R. Larson, B. Liu, and R. Liu, *Operator-valued measures, dilations, and the theory of frames*, *Memoirs Amer. Math. Soc.*, Vol.229 No.1075 (2014).
- [JS15] W. B. Johnson and G. Schechtman, *A Schauder basis for $L_1(0, \infty)$ consisting of non-negative functions*, *Illinois J. Math.*, **59** no. 2 (2015), 337-344.
- [K98] E. Kirkland, *Advanced computing in Electron Microscopy*. Springer. ISBN 978-0-306-45936-8 (1998).
- [K09] P. Kuchment, *Tight frames of exponentially decaying Wannier functions*, *J. Phys. A: Math. Theor.* **42** (2009) 025203.
- [KOPT13] G. Kutyniok, K.A. Okoudjou, F. Philipp, and E.K. Tuley, *Scalable frames*, *Linear Algebra Appl.*, **438** (2013), 2225-2238.
- [KL18] F. Krahmer, Y. Liu, *Phase Retrieval Without Small-Ball Probability Assumptions*, *Information Theory IEEE Transactions on*, **64**, no. 1 (2018), 485-500.
- [MCKS99] J. Miao, P. Charalambous, J. Kirz, and D. Sayre, *Extending the methodology of x-ray crystallography to allow imaging of micromere-sized non-crystalline specimens*, *Nature*. **400** (1999)
- [M12] E. Martin, *Global Coordinate Systems: Continuously Moving Finite-Dimensional Unit Norm Tight Frames on Smooth Manifolds*, Honors Undergraduate Thesis, University of Texas (2012).
- [MP04] , C. Micchelli and M. Pontil, *A function representation for learning in Banach spaces*, in *Learning Theory. Lecture Notes in Computer Science*, vol. 3120 (Springer, Berlin, 2004), 255-269.
- [MSS15] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing Families II: mixed characteristic polynomials and the Kadison-Singer problem*, *Ann. of Math.* **182** (2015), no. 1, 327-350.
- [O79] A. Olevskii, *Completeness in $L_2(\mathbb{R})$ of almost integer translates*, *C. R. Acad. Sci. Paris* **324** (1979), 987–991.

- [OU18] A. Olevskii and A. Ulanovskii, *Discrete translates in function spaces*, Analysis Mathematica, **44** (2018) no 2, 251-261.
- [PS16] A. Powell and A.H. Spaeth, *Nonnegativity constraints for spanning systems*, Transactions of the AMS, **368** (2016), 5783–5806.
- [SB15] M. Speckbacher and P. Balazs, *Reproducing pairs and the continuous non-stationary Gabor transform on LCA groups*, J. Phys. A, Math. Theor. **48**, 395201 (2015).
- [SB17] M. Speckbacher and P. Balazs, *Reproducing pairs and Gabor systems at critical density*, J. Math. Analysis and Appl., Math. Theor. **455** (2017), no. 2, 1072-1087.
- [SB20] M. Speckbacher and P. Balazs, *Frames, their relatives and reproducing kernel Hilbert spaces*, preprint arXiv:1704.02818.
- [SB13] D.T. Stoeva and P. Balazs, *Canonical forms of unconditionally convergent multipliers*, J. Math. Anal. and Appl., **399** (2013) 252-259.
- [T03] G. Taylor, *The phase problem*. Acta Crystallographica Section D. 59 (11): 1881–1890 (2003).
- [V15] C. Vinzant, *A small frame and a certificate of its injectivity*, Sampling Theory and Applications (SampTA) Conference Proceedings. (2015)197-200.
- [Xu17] Z. Xu, *The minimal measurement number for low-rank matrix recovery*, Appl. Comput. Harmon. Anal. (2017)
- [ZHZ09] H. Zhang, Y. Xu, and J. Zhang, *Reproducing kernel Banach spaces for machine learning*, J. Machine Learning Research **10** (2009), 2741-2775.
- [ZZ17] H. Zhang and J. Zhang, *Learning with reproducing kernel Banach spaces*, Conference proceedings, New Trends in Analysis and Interdisciplinary Applications (2017) 417-423.